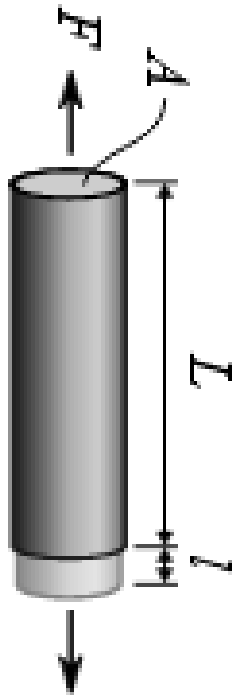


STM Exercise 2



$$\sigma = \frac{F}{A}$$

where

- σ stress [Pa]
- F applied force [N]
- A cross-sectional area [m^2]

$$\varepsilon = \frac{\Delta L}{L_0}$$

where

- ε strain
- ΔL total elongation [m]
- L_0 original length [m]

$$E = \frac{\sigma}{\varepsilon}$$

Exercise 1: You have a cylindrical component made of an Al alloy (10 mm diameter, $Y_S = 150 \text{ MPa}$): which is the maximum load you can apply on this component without having plastic deformation? Consider a safety factor of $n = 2$.

Poisson's ratio, ν

- Poisson's ratio, ν :**

$$\nu = \frac{-\epsilon_x}{\epsilon_z} = \frac{-\epsilon_y}{\epsilon_z}$$

(Isotropic materials only)

metals: $\nu \sim 0.33$

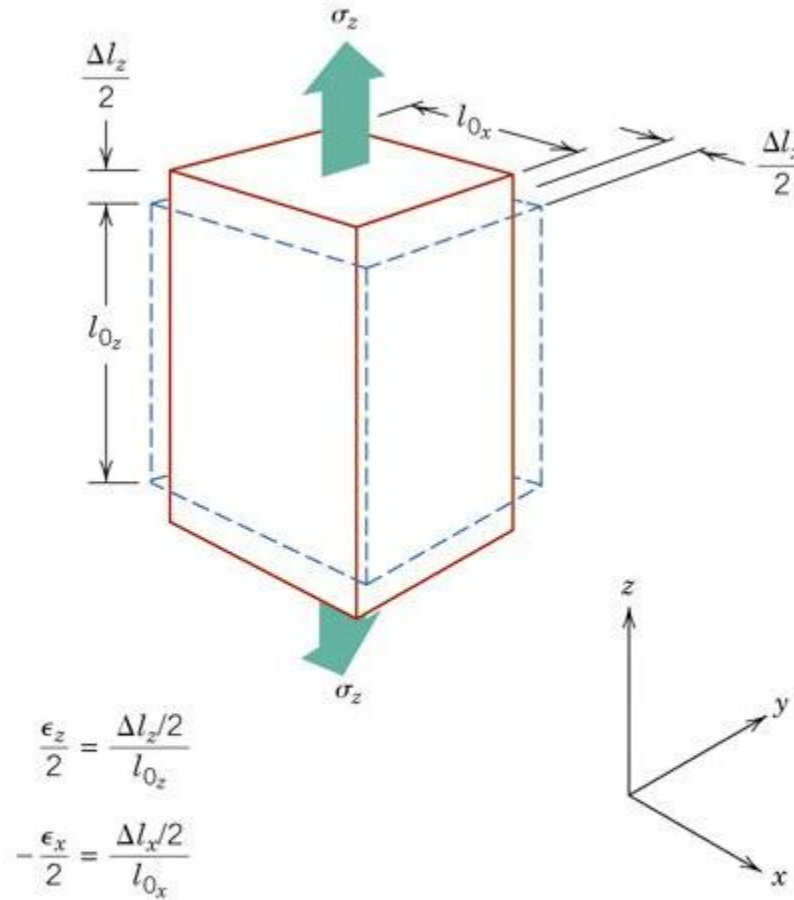
ceramics: $\nu \sim 0.25$

polymers: $\nu \sim 0.40$

Units:

E : [GPa] or [psi]

ν : dimensionless



Exercise 2: You have a cylindrical aluminum component (diameter =10 mm, $\nu=0.34$, $E=70$ GPa): calculate the tensile load you need to apply to have a 2 micron smaller diameter component.

Exercise 3: Which is the diameter you need for a 380 mm long Cu component ($E=110$ GPa) to be loaded at 6600N, if you can accept an elongation of 0.5 mm.

Exercise 4: The elastic modulus of a steel is 193 GPa, of a bronze is 100 GPa, and of a polymer is 0,17 GPa ; calculate the elastic elongation of a 10 meter long wire made of steel, bronze or polymer , cross section is 1 mm², the applied load is F= 10 N.

Exercise 5: A wire 10 m long with a cross section of 1mm^2 underwent tensile loading of 10N. Determine stress, strain and elongation if the wire is made of Copper ($E=110\text{ Gpa}$) or steel ($E=210\text{ Gpa}$). Compare different material on the available excel file.

Exercise 6

Consider an aluminium cylindrical sample with a diameter of 1cm.
the significant values for applied force underlined in traction test are:

- 4600 N (yield),
- 8450 N (maximum)
- 5700 N (rupture) .

When rupture occurs sample diameter is $\frac{1}{3}$ of the initial one

Determine:

- Yield strength
- Tensile strength
- Seemly fracture strength
- True fracture strength